MLR780 Air Conditioning and Refrigeration 2021

Assignment 4
Energy Considerations

Instructions

- This individual assignment is to be submitted online before midnight on **16 May 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment4-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment4-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Outcomes:

This assignment is designed to assist you develop a basic understanding of a VAV system and what the energy implication might be for different operating conditions of the system itself.

Background:

Consider Figure 1 representing a 400-seat lecture hall (500 m^2) which is equipped with a VAV system with terminal re-heat. Figure 2 contains a simplified schematic lay-out of the air circuit under consideration to be used in the analysis. Atmospheric air is drawn in at state 1 and is mixed with return air (red arrows) to produce state 2 before entering a cooling coil (serviced by a chilling plant) which cools and dehumidifies the air to state 3. A centrifugal fan alters the air state to state 4 and enables the distribution of the supply air via supply ducts to the diffusers which are equipped with electric element re-heaters. Some air leakage from the supply duct is possible depending on the quality of construction of the duct system. After possible re-heat, the air is introduced into the lecture hall at state 5. The return air at the return grills are at state 6 (assume a fully mixed ventilation system). An exhaust fan is used to draw the return air through the plenum space above the ceilings.

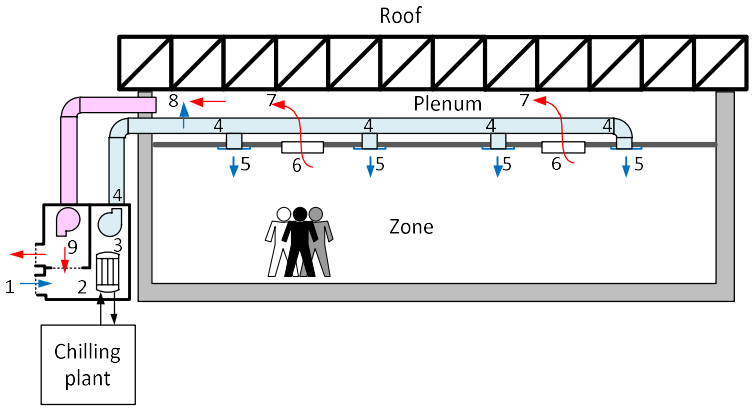


Figure 1 Lecture hall sectional view representation.

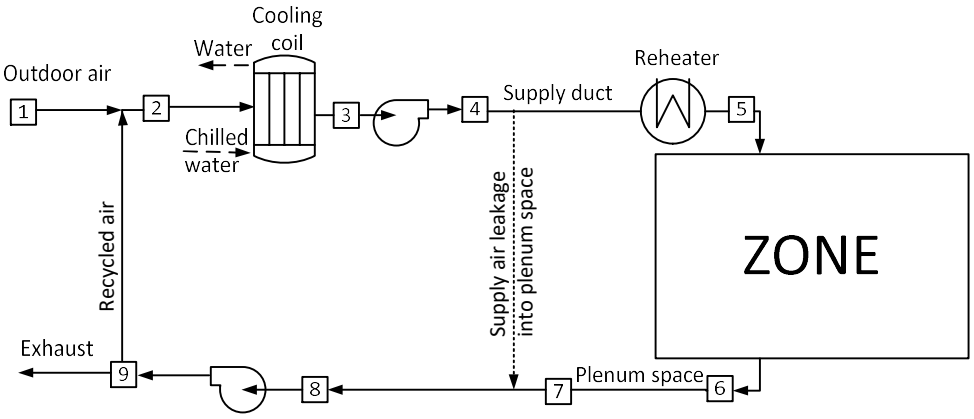


Figure 2 Schematic layout of the air circuit.

The air volume through the cooling coil is directly controlled according to a temperature and humidity feedback loop based on the in-time air condition in the lecture hall. The re-heaters are activated in a similar fashion if needed, but based only on the dry bulb temperature. The following target Psychometric conditions are of interest (use 90 kPa through-out):

- State 1: outdoor air is at 30˚C, 55% relative humidity.
- State 3: set-point of the air exiting the cooling coil is 11˚C (saturated) [1]
- State 6: lecture hall condition is **exactly** 23°C [2], **maximum** 50% relative humidity [3].

Besides the internal cooling loads (occupants), consider the cooling loads supplied in Table 1. Depending on the size of the plenum space and positioning of the return air grills, a significant proportion of the roof cooling load can be intercepted by the return air, which reduces the cooling load in the zone, and possibly the energy consumption

[1] In reality state 3 will not be fixed and is much dependent on the cooling coil and other component performance at different operating conditions. The emphasis of this assignment is not on these.
[2] This strict condition is adopted only for purposes of this assignment. Generally, there will be an acceptable temperature range of the zone (between some minimum and some maximum temperature)
[3] For purposes of this assignment any humidity value less than 50% is acceptable, but generally some minimum value might also be applicable depending on the application.

of the cooling plant. However, if there are significant leakages from the supply duct into the plenum space, the overall energy consumption may increase. Let x (value between 0 and 1) represent the fraction of roof heat intercepted by the return air, and y (value between 0 and 1) represent the fraction of supply air leaking into the plenum.

Table 1 Zone cooling load components.

Component	
Roof	q_{Roof} = 9800 W
Lights/electronics	q_{Lights} = 2500 W
Walls	q_{Wall} = 3800 W

Assume the chilling plant to have a thermal coefficient of performance[4] that is 60% of that of the Carnot coefficient of performance based on states 1 and 3.

Calculations:
Base the specific heat and air density values on the inlet state of any relevant component of air circuit section.

1. Calculate the overall energy consumption [W] (based on the chiller unit and the re-heaters but ignore the energy consumption and efficiency of the fans) for (x,y) = (0.2, 0.1) when the hall is fully occupied. Show all calculations in full. Base the ventilation rates on cooling load requirements as well as minimum ventilation rate requirements. **Do this question for the first mass flow determination iteration as in the class example (IE: state 6 might not be at the intended set-point).**
2. Repeat question 1, but include mass determination convergence, such that state 6 is at the intended set-point.
3. Compare the results from question 1 and 2 in tabular format and discuss were needed:

Symbol		Question 1	Question 2
$\dot{m}_{min,1}$	outdoor air mass flow rate		
$\dot{m}_{s,2}$	mass flow rate to satisfy the sensible load in the zone		
$\dot{m}_{T,2}$	mass flow rate to satisfy the minimum requirement of the total load in the zone		
t_6	Actual zone temperature		
ϕ_6	Actual zone relative humidity		
q_{CC}	Cooling coil		
q_{RH}	Reheater		
q_{CH}	Chiller		

[4] Chiller plant efficiency (atmosphere air to supply air) is generally complicated and depend on many component interactions and energy transfer mechanism. For calculation purposes a simplification of this efficiency is adopted here.

Consider two occupancy levels of the lecture hall: Case A: Fully occupied, Case B empty.

4. Repeat the calculation of question 2 for other combinations of x and y (given below). Tabulate your answers for Cases A and B separately as follows (Do not show detailed calculations, just the final converged answers):

Overall energy consumption [W]		x		
		0	0.3	0.6
y	0			
	0.1			
	0.2			

5. Comment on your answers in terms of the impact of occupancy, roof heat absorption and air leakage. Make three-dimensional plots of the energy consumption in terms of x and y for Cases A and B to support your conclusions, and a give bar charts to compare t_6, ϕ_6, q_{CC}, q_{RH} and q_{CH}.

INDICATE HOW MUCH TIME YOU SPENT ON THIS ASSIGNMENT